

TAPAN KHANDELWAL

+91 9982866695 khandelwaltapan7@gmail.com
linkedin.com/in/tapan-khandelwal github.com/KhandelwalTapan7

Profile Summary

AI Engineer with AWS Certified AI Practitioner and Cloud Practitioner certifications, specializing in Generative AI, Agentic AI, and LLM-powered systems. Experienced in building multi-agent pipelines and RAG applications with LangChain and LangGraph, integrating LLM providers such as OpenAI, Gemini, Groq, and Mistral AI, and deploying cloud-native solutions on AWS (Lambda, EC2, API Gateway, Bedrock, SageMaker). First-author researcher on deepfake detection using a hybrid CNN-LSTM model, published in IJCSE (2026). Improved a production text classification pipeline's weighted F1 from 0.71 to 0.86 at Cognus Technology. Seeking an AI Engineer or Generative AI Developer role.

Education

Jagannath University

B.Tech in Computer Science (AI/ML) – CGPA: 8.2

Jaipur, India

Sep 2022 – June 2026

Research Publication

- **Aurora: A Hybrid Deep Learning Approach for Deepfake Detection and Digital Content Authentication** – first author, *IJCSE*, Vol. 14, Issue 9, Sep 2026 (DOI: 10.26438/ijcse.v14i9.7516). Built a CNN-LSTM detector with hash-based integrity checks, deployed as a real-time web app; 95.2% validation accuracy on 84,377 images.

AI/ML Projects

Multi-Agent AI Trip Planner

Tech: Python, LangGraph, Groq, AWS Lambda, Cognito, DynamoDB

ai-trip-planner-multi-agent-2.onrender.com

- Built a 5-agent LangGraph system (planner, budget, local info, transport, writer) with parallel fan-out/fan-in, where three agents run concurrently before merging into one itinerary.
- Deployed three ways on AWS (serverless, Docker on EC2, load-balanced EC2) with JWT auth via Cognito and manual JWKS verification, using async write-then-poll to avoid API Gateway's 30-second timeout.

DocSight AI – Multimodal Agentic RAG Assistant

Tech: Python, FastAPI, LangGraph, Qdrant, Groq, Chainlit

docsight-ai-fyn3.onrender.com/app/

- Built a multimodal, agentic RAG pipeline (FastAPI + LangGraph ReAct agent) letting users upload any PDF and receive cited, page-level answers, with vision-based extraction of charts and diagrams.
- Engineered per-session document isolation with Qdrant Cloud, deploying the full stack (API + Chainlit UI) as one containerized service on Render.

Professional Experience

AI/ML Engineer

June 2025 – May 2026

Cognus Technology

Jaipur

- Built a Python NLP text classification pipeline (scikit-learn, TF-IDF) on 3,000 labeled support tickets across 6 categories, lifting weighted F1 from 0.71 to 0.86, then shipped it as a deployable script using Git.
- Automated triage by auto-routing high-confidence predictions and escalating low-confidence ones for human review, cutting manual triage volume by 25 to 30 percent.

AI/ML Engineer Intern

July 2024 – Aug 2024

iGeeks Technology

Bangalore

- Cleaned and prepared a 5,000-row dataset in Python/SQL, handling missing values, outliers, and categorical encoding ahead of model training.
- Ran feature engineering and 5-fold cross-validation across Logistic Regression, Random Forest, and Gradient Boosting, improving validation accuracy from 71 to 89 percent, and built automated logging to flag production data drift.

Skills

LLMs & GenAI: OpenAI, Gemini, Groq, Mistral AI, Ollama, Claude, LangChain, LangGraph, Multi-Agent Systems, MCP, Tool Calling

RAG & Vector Search: RAG, Chroma, Qdrant, Pinecone, Sentence-Transformers, RAG Evaluation

DL, CV & NLP: PyTorch, TensorFlow, CNN-LSTM, OpenCV, Hugging Face Transformers, NLTK, Text Classification

Backend & Programming: Python, Java, C++, FastAPI, Flask, REST APIs, JWT/JWKS Auth, boto3, n8n

Data & ML: SQL, MySQL, PostgreSQL, MongoDB, DynamoDB, Scikit-learn, Cross-Validation, EDA, Power BI, Tableau

Cloud & DevOps: AWS (EC2, S3, Lambda, API Gateway, IAM, Cognito, SageMaker, Bedrock), Docker, Git, GitHub

Certifications

- **AWS Certified AI Practitioner** – Amazon Web Services, Sep 2026 (ID: 13e2f2002bff4c63b023c47935823ae9)
- **AWS Certified Cloud Practitioner** – Amazon Web Services, Sep 2026 (ID: b876ec05c391414b8586b4f52c03837b)
- **Networking Essentials** – Cisco Networking Academy